$$\frac{\Delta f_1}{\Delta f_2} = \frac{m_m \Delta r_1}{r_1^3} \cdot \frac{r_2^3}{m_s \Delta r_2} = \frac{m_m}{m_s} \cdot \frac{r_2^3}{r_1^3} \cdot \frac{\Delta r_1}{\Delta r_2}$$

using $\Delta r_1 = \Delta r_2 = 2R_{earth} \quad m = 8 \times 10^{22} kg, M_s = 2 \times 10^{30} kg$

$r_1 = 0.4 \times 10^6 km, r_2 = 150 \times 10^6 km$

we get $\frac{\Delta f_1}{\Delta f_2} = 2$

Q. 9 thin uniform tube is bent into a circle of radius r in the vertical plane. Equal volumes of two immiscible liquids, whose densities are p_1 and $p_2 (p_1 > p_2)$, fill half the circle. The angle θ between the radius vector passing through the common interface and the vertical is :

Option 1:

$\theta = \tan^{-1} \pi \left(\frac{P_1}{P_2} \right)$

Option 2:

$\theta = \tan^{-1} \frac{\pi}{2} \left(\frac{P_2}{P_1} \right)$

Option 3:

$\theta = \tan^{-1} \left[\left(\frac{P_1 - P_2}{P_1 + P_2} \right) \right]$

Option 4:

$\theta = \tan^{-1} \frac{\pi}{2} \left[\left(\frac{P_1 + P_2}{P_1 - P_2} \right) \right]$

Correct Answer:

$\theta = \tan^{-1} \left[\left(\frac{P_1 - P_2}{P_1 + P_2} \right) \right]$

Solution:
As we learned

Absolute Pressure -

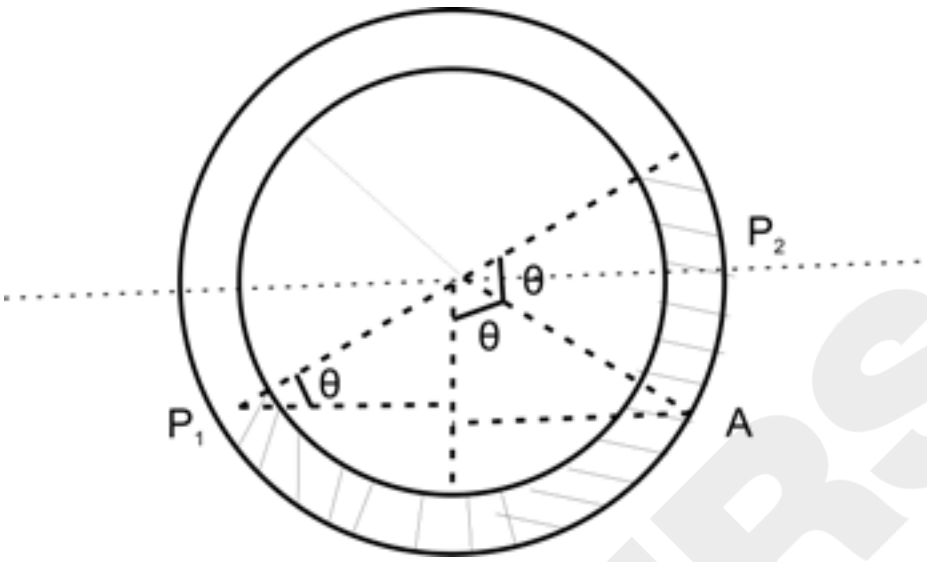
$P = P_0 + \rho g h$

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- wherein

$P \rightarrow hydrostatics\ Pressure$

$P_0 \rightarrow atmospheric\ Pressure$



equating presure at point A

$$p_1 gR(\cos\theta - \sin\theta) = p_2 gR(\sin\theta + \cos\theta)$$

$$\frac{p_1}{p_2} = \frac{\sin\theta + \cos\theta}{\cos\theta - \sin\theta} = \frac{\tan\theta + 1}{1 - \tan\theta}$$

$$p_1 - p_1 \tan\theta = p_2 + p_2 \tan\theta$$

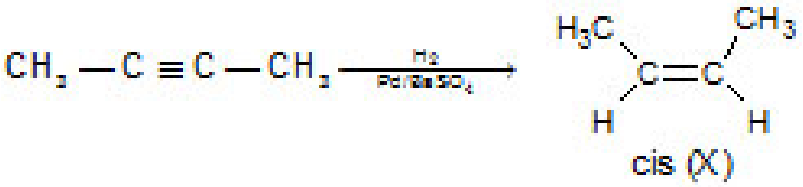
$$\Rightarrow (p_1 + p_2)\tan\theta = p_1 - p_2$$

$$\Rightarrow \tan\theta = \frac{p_1 - p_2}{p_1 + +p_2} \Rightarrow \theta = \tan^{-1}\left(\frac{p_1 - p_2}{p_1 + p_2}\right)$$

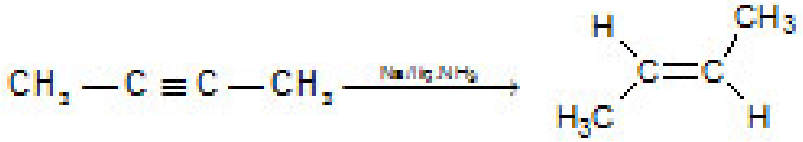
Q. 10 A Carnot's engine works as a refrigerator between 250 K and 300 K. It receives 500 cal heat from the reservoir at the lower temperature. The amount of work done (in J) in each cycle to operate the refrigerator is :

Correct Answer:
420

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cis (X)

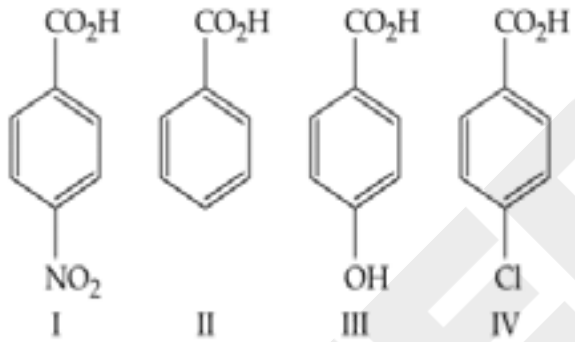


Trans (Y)

Cis has higher dipole-moment than trans.

Therefore, **option (1) is correct.**

Q. 24 The increasing order of the acidity of the following carboxylic acids is:



I II III IV

Option 1:
I < III < II < IV

Option 2:
IV < II < III < I

Option 3:
II < IV < III < I

Option 4:
III < II < IV < I

Correct Answer:
III < II < IV < I

Solution:
As we learnt in

Acidic strength of carboxylic acid -

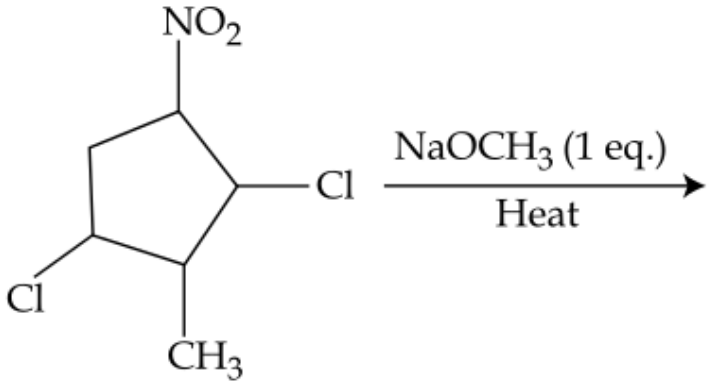
-NO_2 and -Cl are electron withdrawing groups while -OH is electron donating group.

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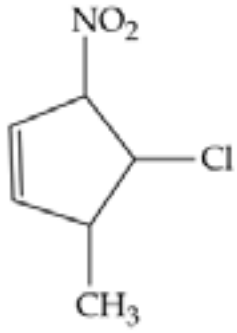
Ring deactivating groups increases the acidic character whereas ring activating groups decreases the acidic strength.

Therefore, **Option(4) is correct.**

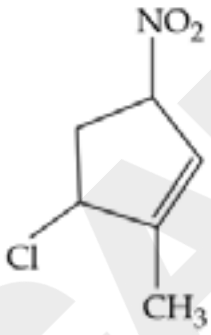
Q. 25 The major product formed in the following reaction is :



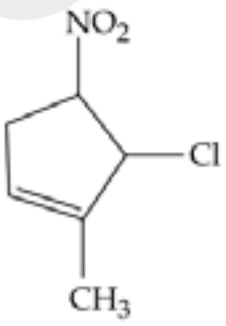
Option 1:



Option 2:



Option 3:



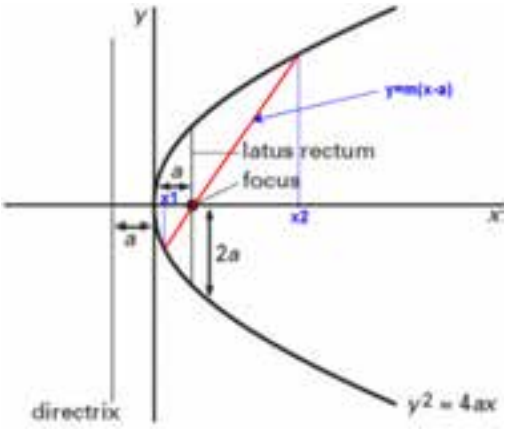
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As we learned,

Standard equation of parabola -

$y^2 = 4ax$

- wherein



Equation of COC PQ is

T = 0

$T \equiv 4(x + x_1) - yy_1 = 0$

Where (x_1, y_1) is $(-8, 0)$

Chord of contact is x = 8

P(8,8) and Q(8,-8)

focus = (2,0)

$\triangle PQF = \frac{1}{2}(8 - 2) \times (8 + 8) = 48$ sq units.

Q. 22 A normal to the hyperbola, $4x^2 - 9y^2 = 36$ meets the co-ordinate axes x and y at A and B, respectively. If the parallelogram OABP (O being the origin) is formed, then the locus of P is :

Option 1:

$4x^2 + 9y^2 = 121$

Option 2:

$9x^2 + 4y^2 = 169$

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Option 3:
$4x^2 - 9y^2 = 121$

Option 4:
$9x^2 - 4y^2 = 169$

Correct Answer:
$9x^2 - 4y^2 = 169$

Solution:
Given Hyperbola $\Rightarrow$ $4x^2 - 9y^2 = 36$

$a^2 = 9, b^2 = 4$

(x_o, y_o) be point on the hyperbola where the normal is drawn

Equation of normal at this point (using point form) is

$$\frac{a^2 x}{x_o} + \frac{b^2 y}{y_o} = a^2 + b^2$$

$$\frac{9x}{x_o} + \frac{4y}{y_o} = 13$$

If this line intersects X axis at A and Y-axis at B

Put y = 0 and x = 0 respectively

$$A = \left(\frac{13x_o}{9}, 0\right), B\left(0, \frac{13y_o}{4}\right)$$

Let P be (x,y)

Midpoint of OB = $B\left(0, \frac{13y_o}{8}\right)$

Midpoint of AP = $\left(\frac{\frac{13x_o}{9} + x}{2}, \frac{y}{2}\right)$

As these mid-points co-incide

Thus $x_o = -\frac{9x}{13}, \ y = \frac{4y}{13}$ .....(i)

Now as (x_o, y_o) lie on given hyperbola

$4x_o^2 - 9y_o^2 = 36$

Putting values in terms of x and y from (i)

$9x^2 - 4y^2 = 169$

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Q. 23 An angle between the lines whose direction cosines are given by the equations, $l+3m+5n=0$ and $5lm-2mn+6nl=0$, is :

Option 1:

$\cos^{-1}\left(\frac{1}{3}\right)$

Option 2:

$\cos^{-1}\left(\frac{1}{4}\right)$

Option 3:

$\cos^{-1}\left(\frac{1}{6}\right)$

Option 4:

$\cos^{-1}\left(\frac{1}{8}\right)$

Correct Answer:

$\cos^{-1}\left(\frac{1}{6}\right)$

Solution:
As we learned,

Direction Cosines -

If α, β, γ are the angles which a vector makes with positive X-axis,Y-axis and Z-axis respectively then $\cos\alpha, \cos\beta, \cos\gamma$ are known as diresction cosines, generally denoted by (l,m,n).

$l=\cos\alpha, m=\cos\beta, n=\cos\gamma$

$l^2+m^2+n^2=1$

$\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

- wherein

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